

ISA 419: Data-Driven Security

21: Touchalytics Lab

Fadel M. Megahed, PhD

Professor
Farmer School of Business
Miami University

 @FadelMegahed

 fmegahed

 fmegahed@miamioh.edu

 Automated Scheduler for Office Hours

Spring 2025

Learning Objectives for Today's Class

- Examine an Android Phone's behavioral touch feature dataset
- Replicate the analysis performed by a InfoSec research paper
- Report the code and results in an understandable manner to a general audience

An Overview of the Touchalytics Paper

The Touchalytics Paper

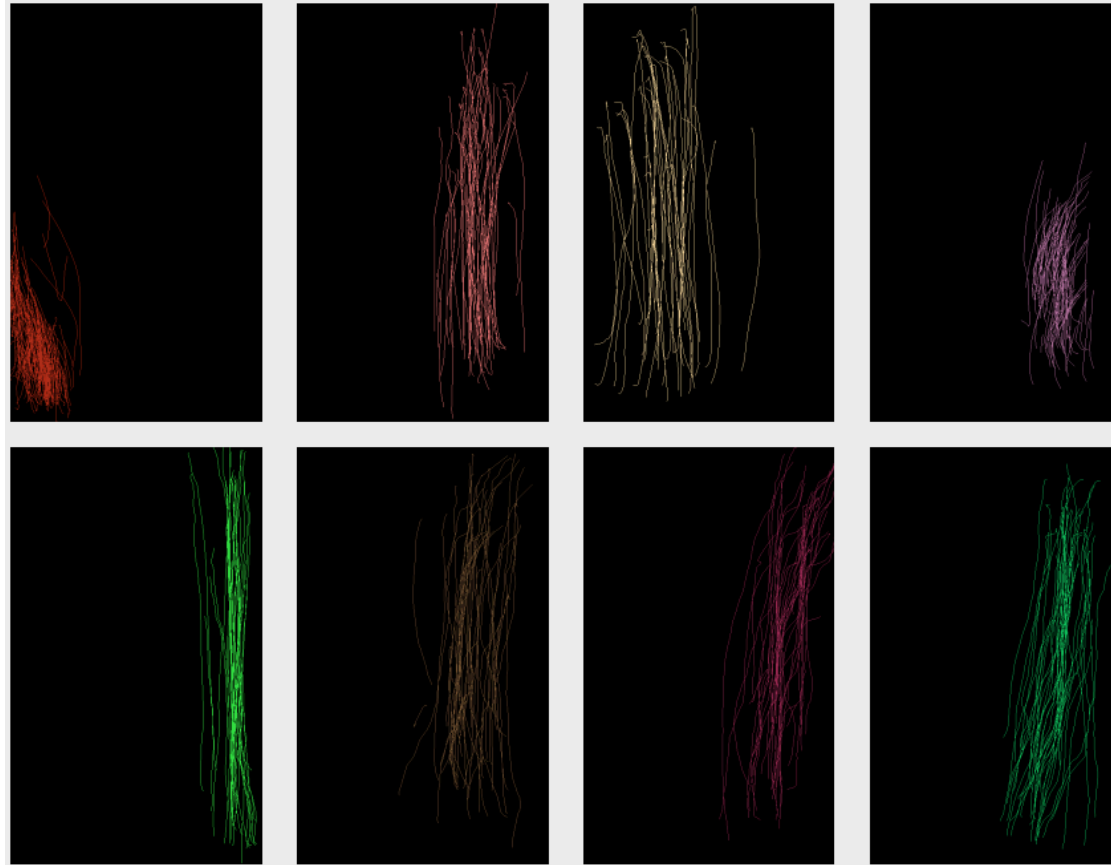
This paper is one of the seminal papers, where smart phone touch features are engineered and used for continuous authentication. The paper has over 1,000 citations, and provides a website that contains:

- The raw data (do not forget to examine the readme file)
- The extracted features (see the readme)
- Matlab Script File for how the features were engineered from the raw data file
- PDFs of their published work; for our lab, we are interested in the Touchalytics Paper

Paper's Motivation [1]

- There is limited work on continuous authentication for touchscreen devices
- Use of smart phone devices typically involves "atomic navigation behavior", which consists of simple and short movements
- Is it possible to authenticate users while they perform basic navigation steps on a touchscreen device and without any dedicated and explicit security action that requires attention from the user?

Paper's Motivation [2]



Strokes from eight different users, each reading three different texts on an Android Phone

Paper's General Idea

There are two phases for learning and classifying touch behavior.

- **Enrollment Phase:** Phase where the system relies on a conventional authentication approach (e.g., password). The authors distinguish between two different "trigger-actions":
 - Sliding horizontally over the screen; for example, to browse through images
 - Sliding vertically over the screen to move content up and down
 - Note that the authors decided to use only on on **single touch gestures**
- **Continuous Authentication Phase:** During this phase, the system continuously tracks all strokes and the classifier estimates if they were made by the legitimate user. For t consecutive negative classification results, the system resorts back to the initial entry-point based authentication method and challenges the user.

Paper's Goals

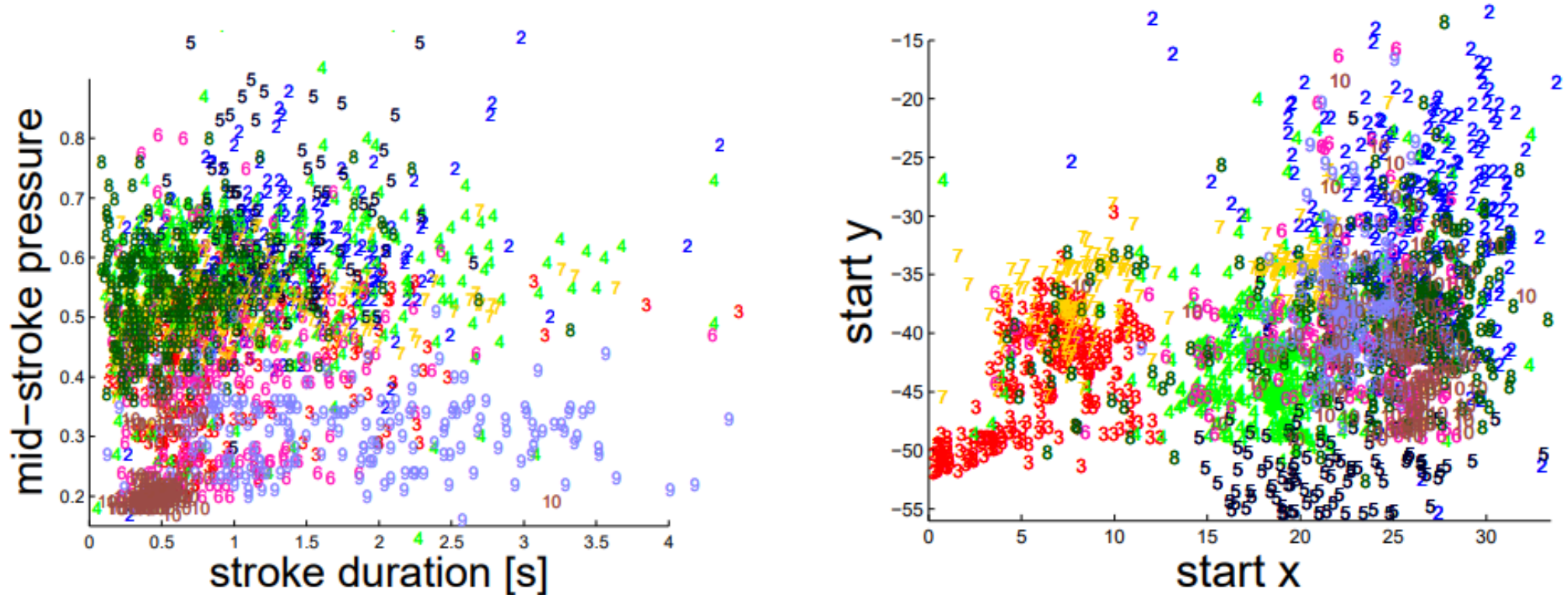
The main goal of their study is to analyze how robustly our proposed framework can distinguish users from each other.

- What is the probability of rejecting a legitimate user?
- What is the probability of accepting an attacker?
- How long does the classifier need to make an authentication decision?
- How robust is the classification within one session, across multiple sessions, and after one week?

Experimental Protocol

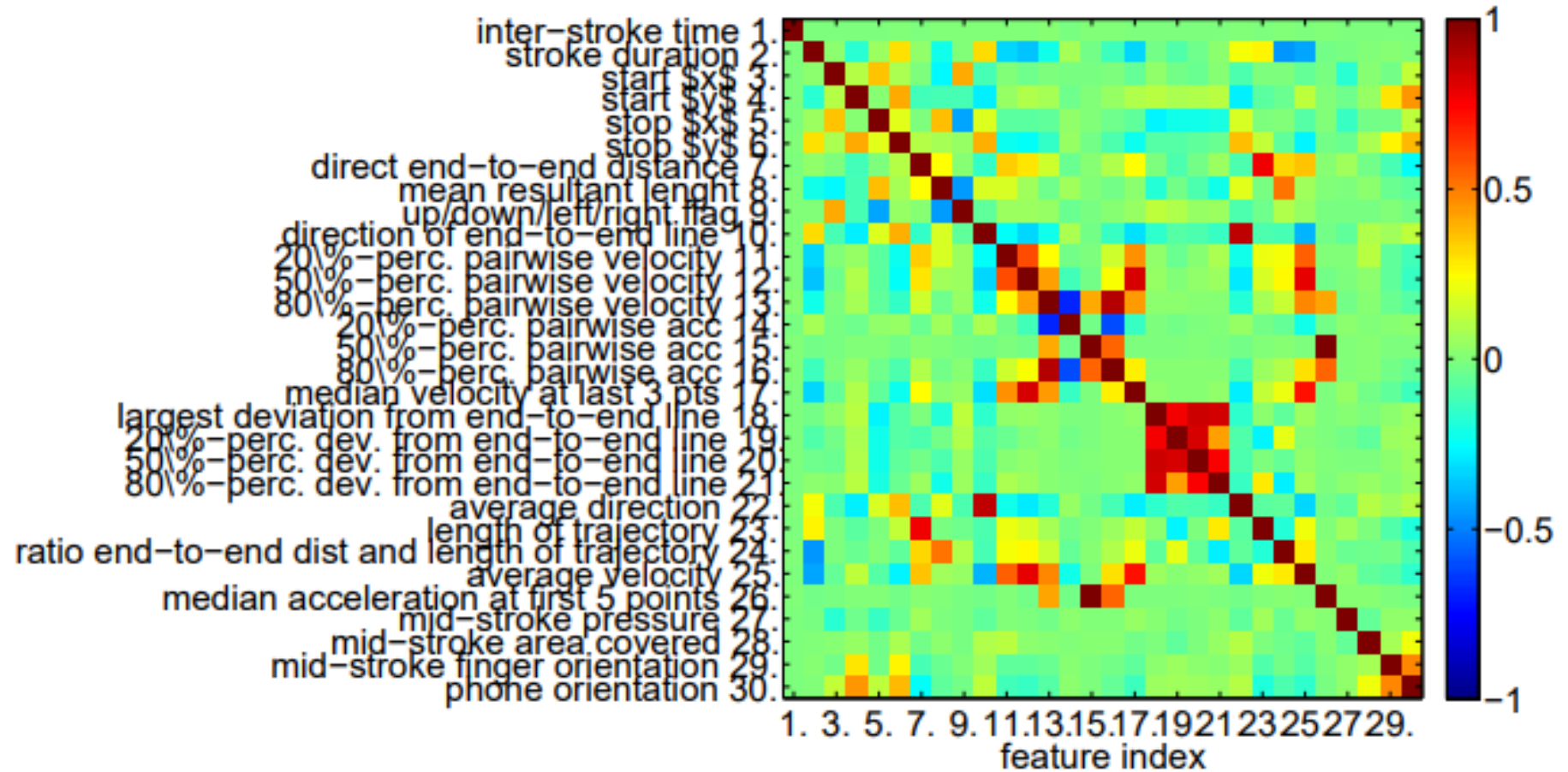
- Sessions 1-3 occurred during the same day and separated by a questionnaire. This is followed by a second phase, where users were asked to spot differences between images.
- Sessions 6-7 occurred one week post the sessions in the previous bullet.

Touch Analytics [1]



Stroke features projected on a 2D-subspace. The user ID is given as a colored number

Touch Analytics [2]



Correlation among the 30 features

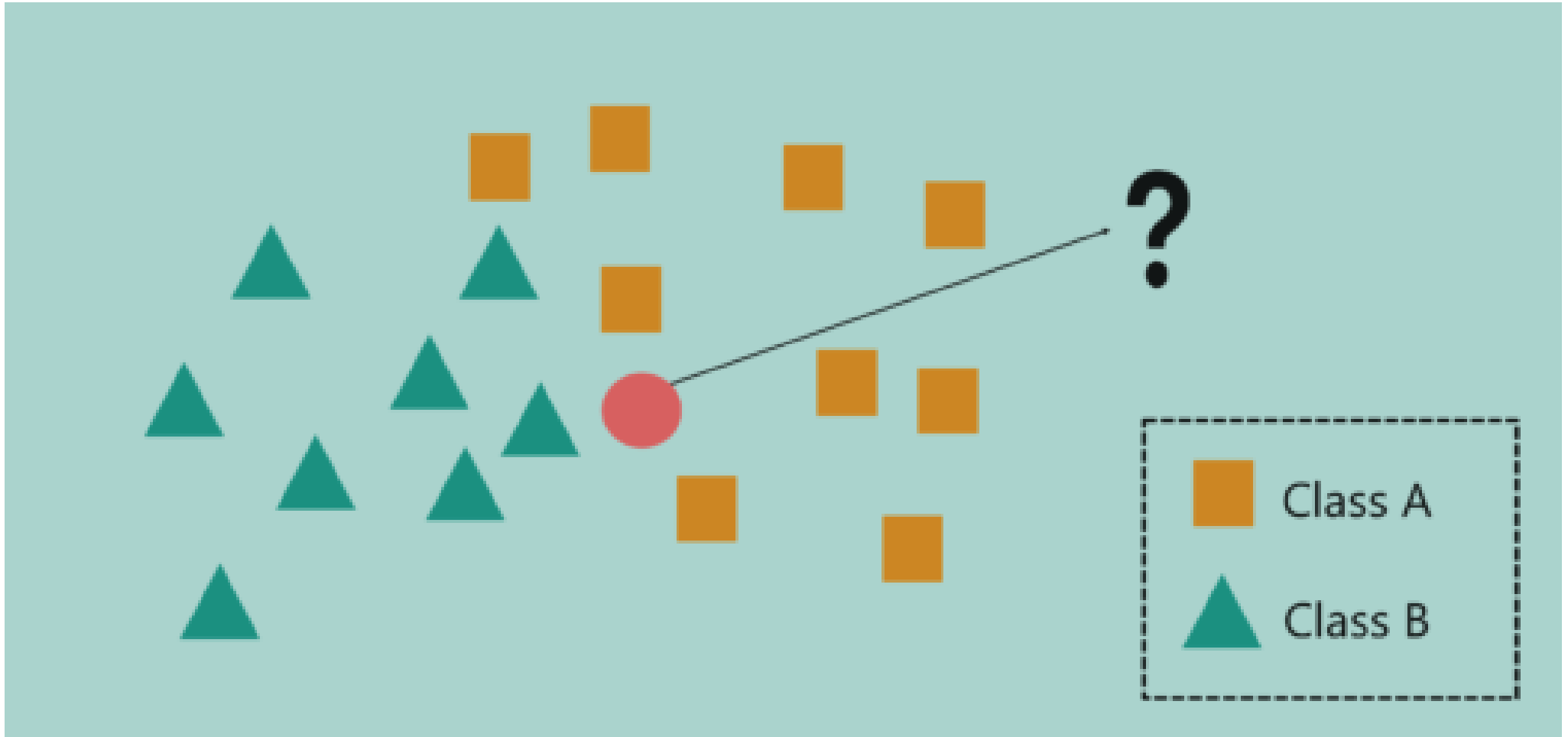
Classification Schemes: kNN

The kNN (k-Nearest Neighbors) algorithm has the following features:

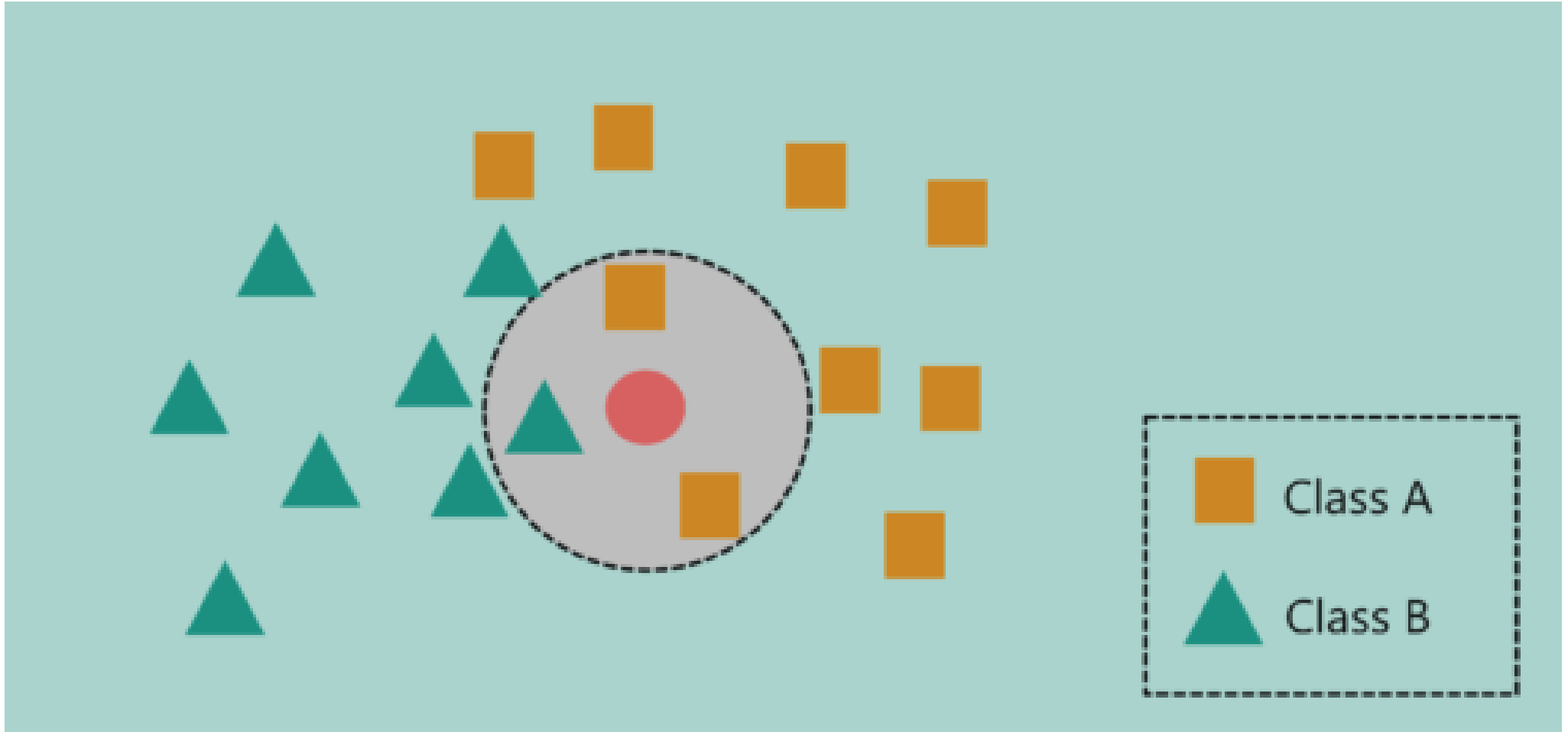
- Supervised learning algorithm
- Utilizes feature similarity
- Lazy algorithm since it uses the entire training data to make a decision instead of coming up with a discriminative function

The following kNN Slides are based on the [edureka.co Blog](#).

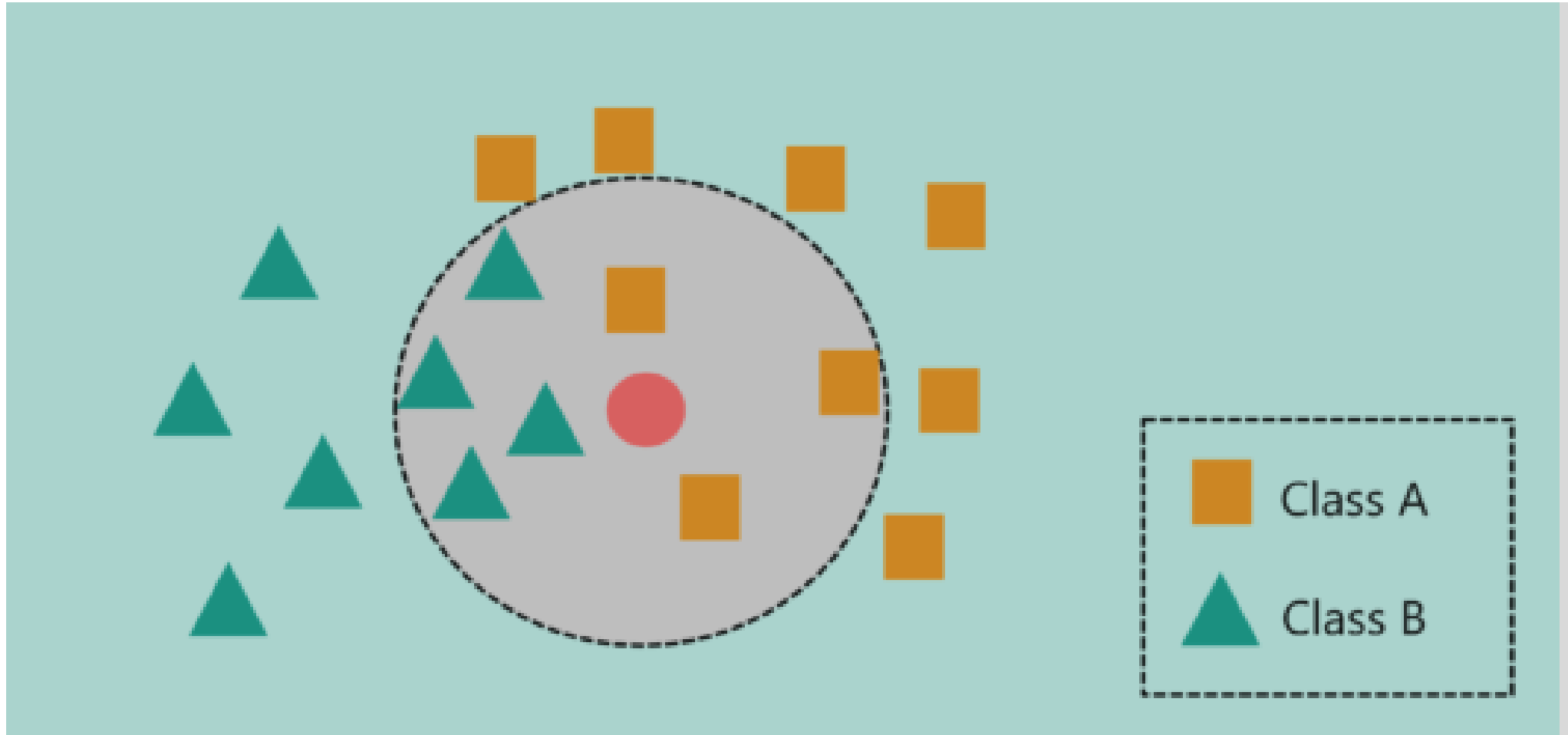
Classification Schemes: kNN



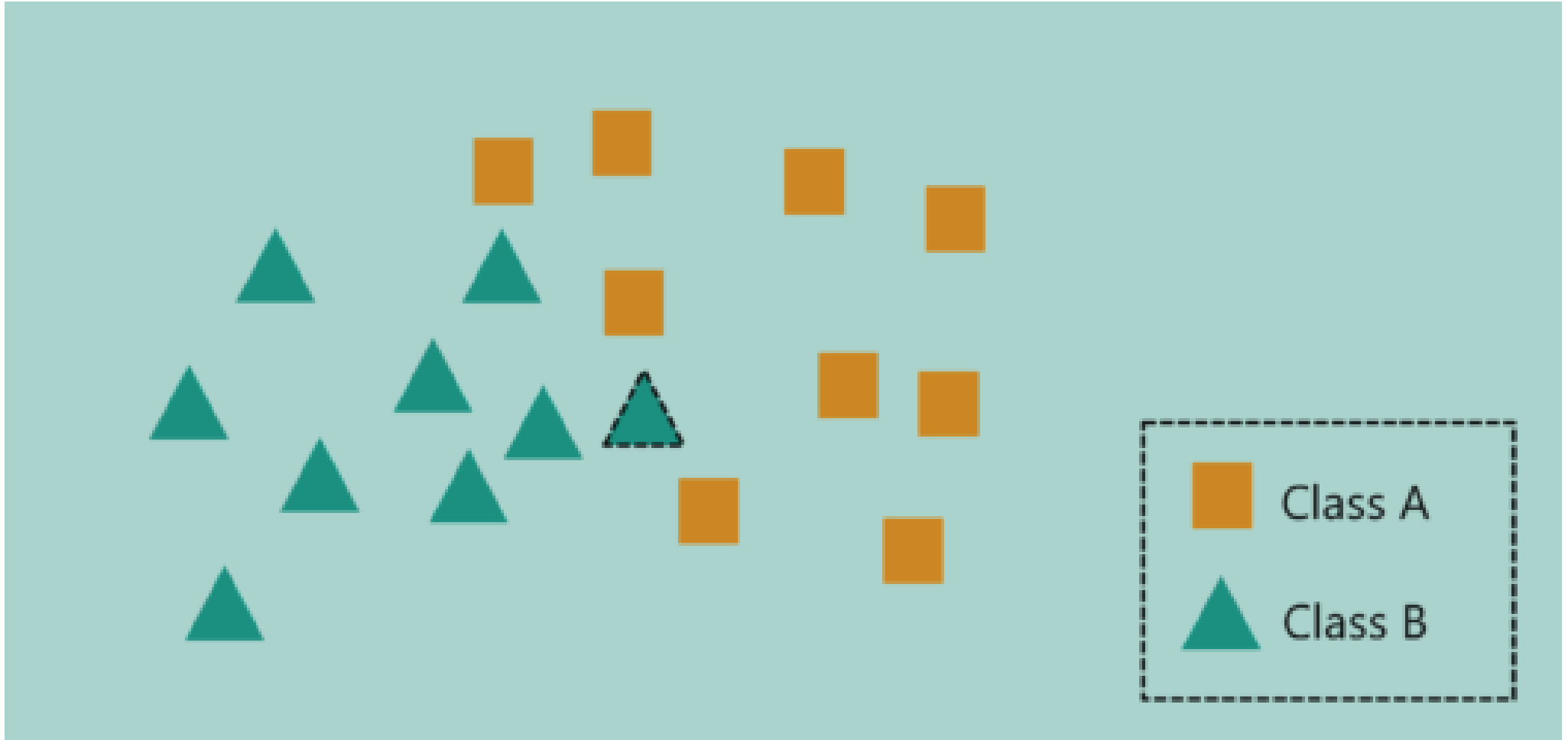
Classification Schemes: kNN



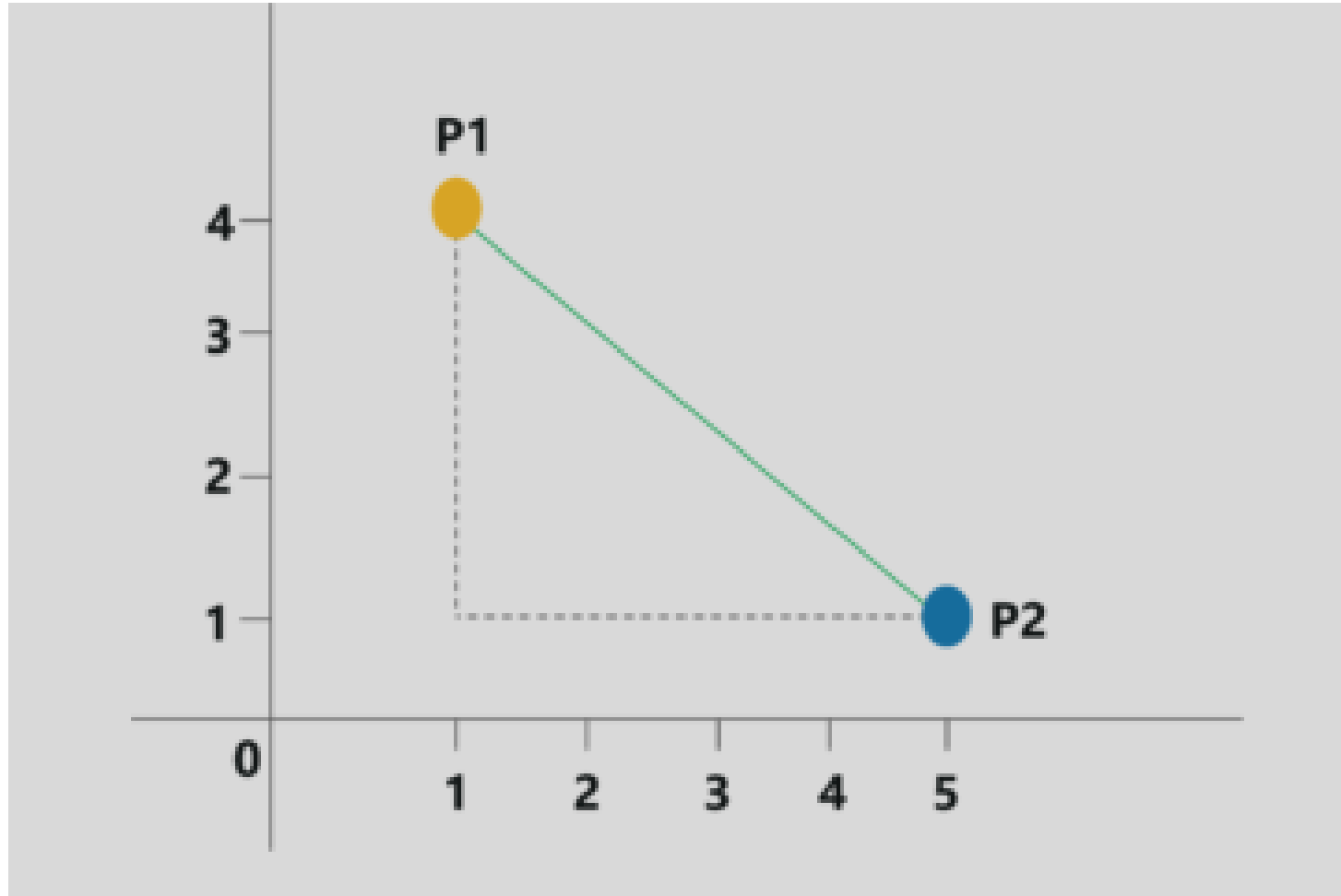
Classification Schemes: kNN



Classification Schemes: kNN



Classification Schemes: kNN



Classification Schemes: kNN

Point P1 = (1,4)

Point P2 = (5,1)

$$\text{Euclidian distance} = \sqrt{(5 - 1)^2 + (4 - 1)^2} = 5$$

Classification Schemes: kNN

Pseudo Code

- Calculate $D(x, x_i)$, where $i = 1, 2, \dots, n$ and D is the Euclidean measure between the data points.
- The calculated Euclidean distances must be arranged in ascending order.
- Initialize k and take the first k distances from the sorted list.
- Figure out the k points for the respective k distances.
- Calculate k_i , which indicates the number of data points belonging to the i th class among k points i.e. $k \geq 0$.
- If $k_i > k_j \forall i \neq j$; put x in class i .

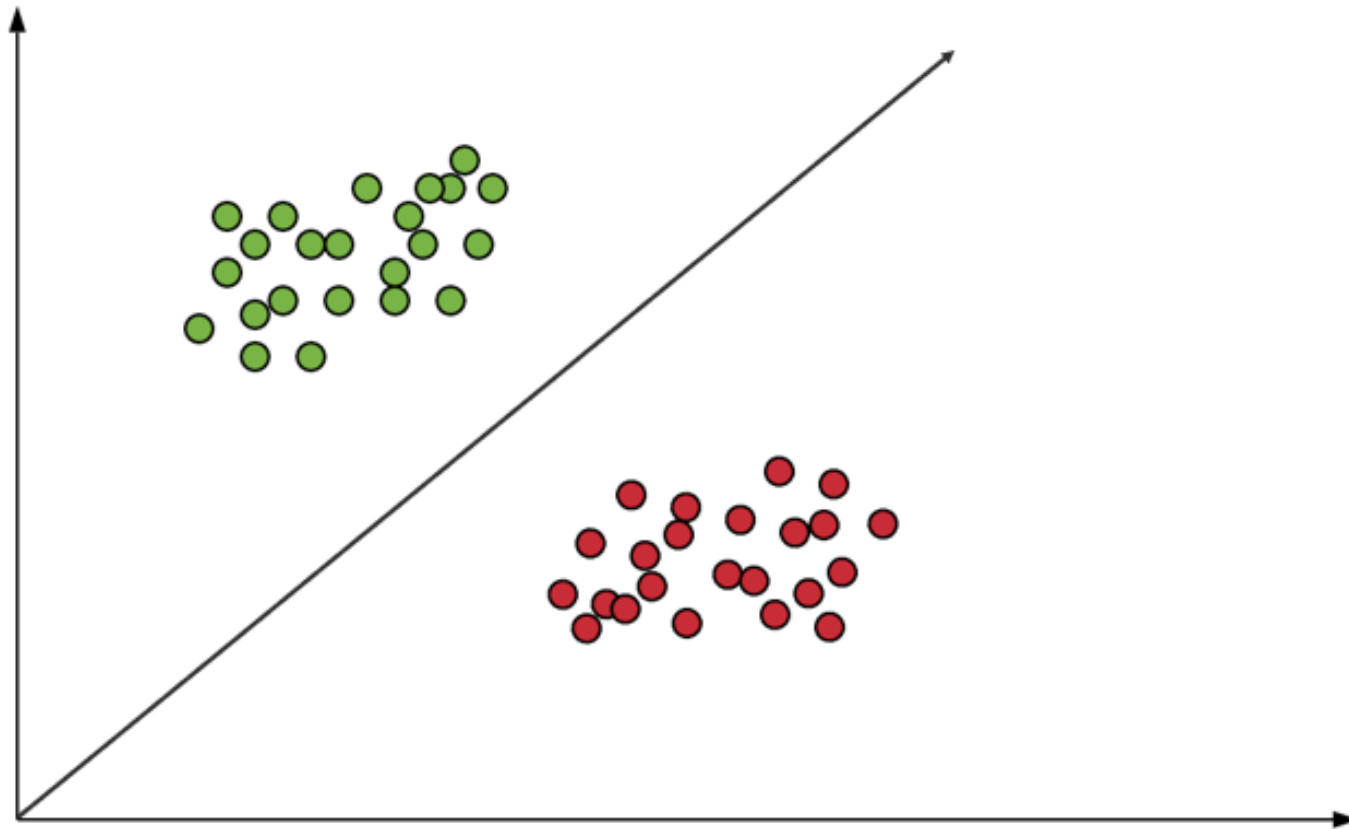
Note that in the paper, they have used $k \in [1, 3, 5, 7]$ to avoid ties.

Classification Schemes: SVM

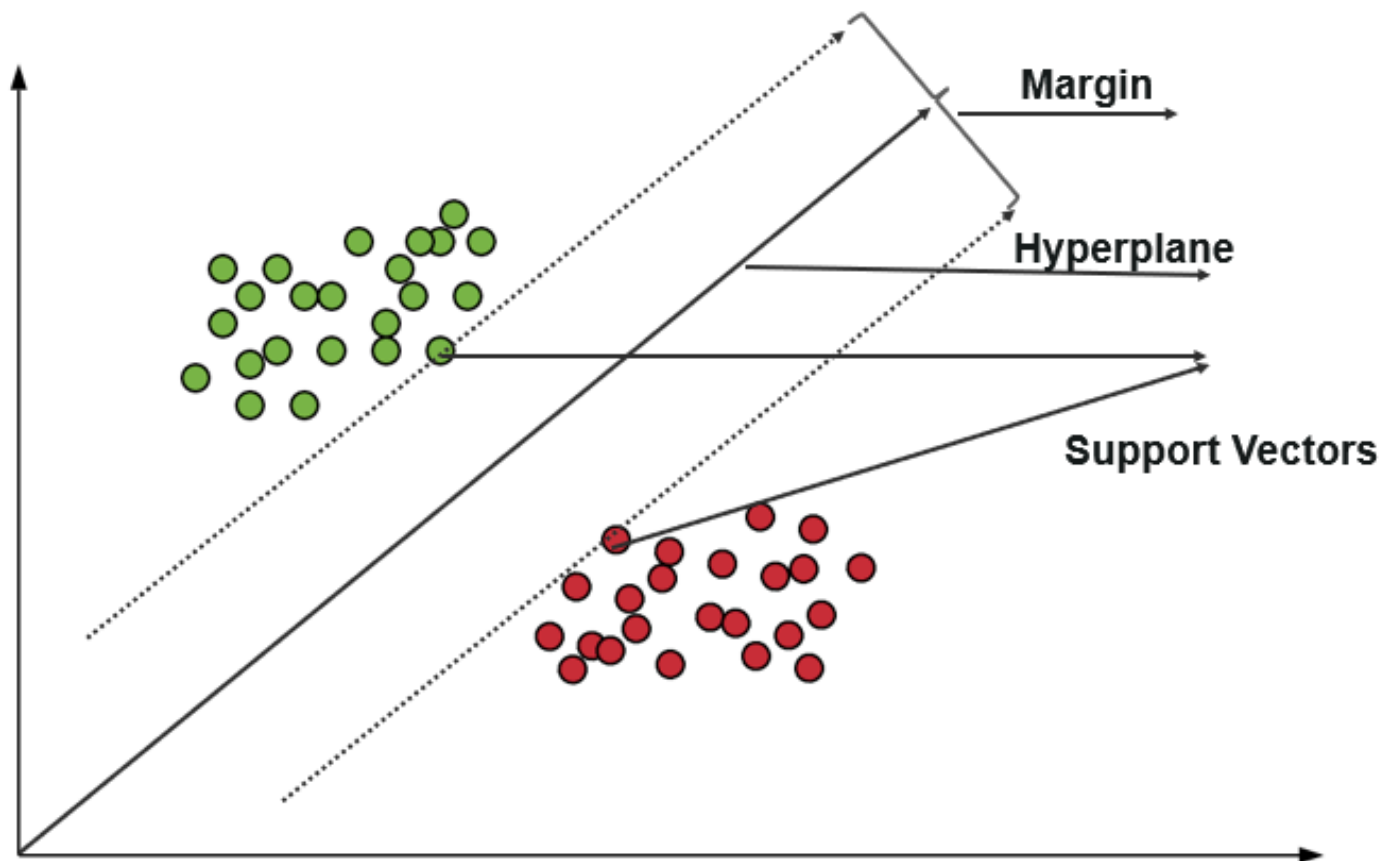
In this paper, they used it for binary classification. They used a RBF for the kernel, which means that they assumed that the data is not linearly separable.

The following SVM Slides are based on the [edureka.co Blog](#).

Classification Schemes: SVM



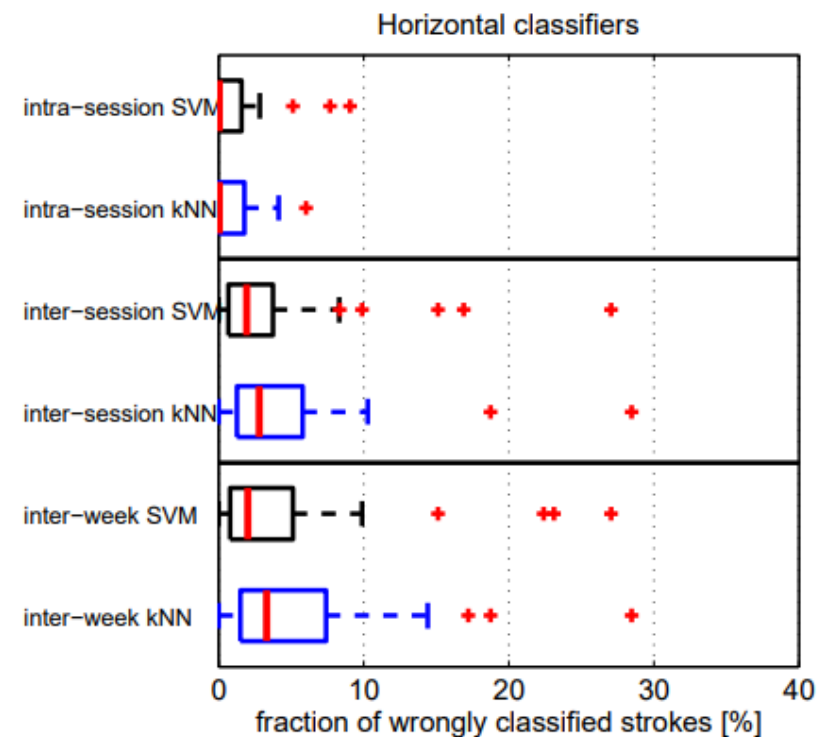
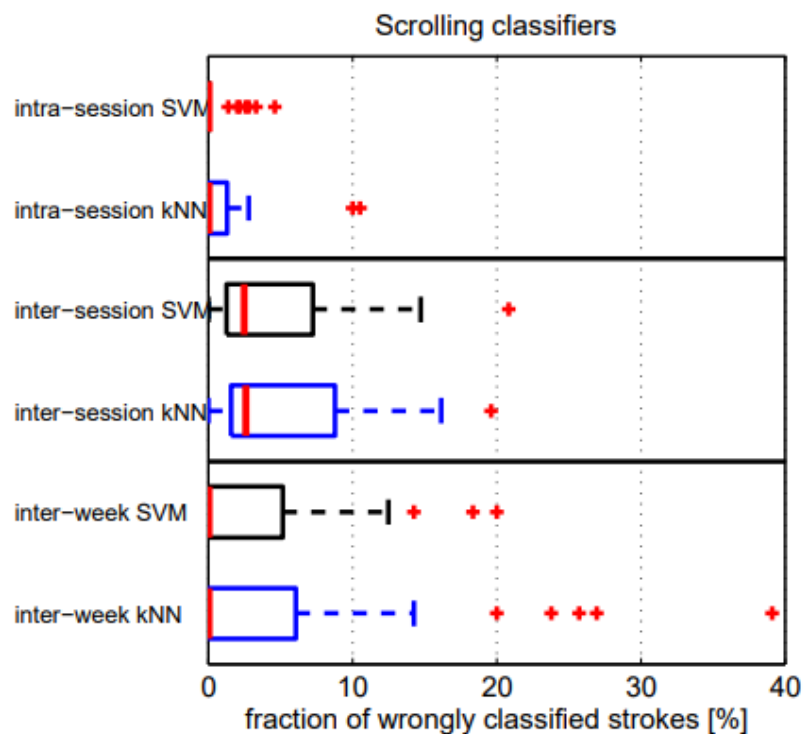
Classification Schemes: SVM



Lab Instructions

Lab Instructions

For the purposes of Lab 04, I would like you to replicate the analysis performed by @frank2012touchalytics. You can start with the [The extracted features](#) and simplify the problem into a binary classification problem, where you will need to recode the data at least 40 times such that for a given user we compare him/her to everyone else.



Lab Instructions

Rubric:

- A student will score 80% on the assignment if they can successfully use both kNN and SVM to distinguish one user of their choice from the rest of the group based on the **inter-week** sessions.
- A student will score 100% on the assignment if they can successfully use both kNN and SVM to distinguish one user of their choice from the rest of the group based on two different training approaches: (a) intra-session, (b) inter-session, and/or (c) inter-week sessions.
- A student will score 120% on the assignment if they can successfully use both kNN and SVM to distinguish all users from the rest of the group based on all three different training approaches: (a) intra-session, (b) inter-session, and (c) inter-week sessions.

You can work individually or in a group of two. Your submission should include: (a) your Notebook, and (b) a CSV file with results from your code.

Recap

Summary of Main Points

By now, you should be able to do the following:

- Examine an Android Phone's behavioral touch feature dataset
- Replicate the analysis performed by a InfoSec research paper
- Report the code and results in an understandable manner to a general audience



Review and Clarification



- **Class Notes:** Take some time to revisit your class notes for key insights and concepts.
- **Zoom Recording:** The recording of today's class will be made available on Canvas approximately 3-4 hours after the end of class.
- **Questions:** Please don't hesitate to ask for clarification on any topics discussed in class. It's crucial not to let questions accumulate.